Shivani Nandakumar

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EDUCATION

University of California, Riverside *Riverside, CA*
B.S. in Computer Science *Graduated: Dec 2025*
GPA: 3.86 / 4.00

M.S. in Computer Science *Expected Graduation: March 2027*

Relevant Coursework: Artificial Intelligence, Natural Language Processing, Information Retrieval, Database Management Systems, GPU Programming, Operating Systems, Computer Networks, Software Engineering, Design & Architecture of Computer Systems

EXPERIENCE

Delta Air Lines *Atlanta, GA (Hybrid)*
Technical Operations Co-Op May 2024 – Present

- Developing and maintaining the **MRO Components Capability List**, a large-scale enterprise tracking system for aircraft part certifications, serviceability data, and component repair capabilities across multiple fleets.
- Building a **digital Bill of Materials (BOM)** for the **LEAP engine** by parsing complex XML files, filtering component data, and automating hierarchy generation with **Python using Pandas, BeautifulSoup, numpy, scikit-learn** to apply AI automation.
- Wrote Python scripts to analyze part interdependencies, identify missing records, and visualize component lineage across thousands of serialized parts.
- Continuing across two additional rotations to expand automation and data-driven decision support systems.

LifeStages *Remote*
Software Engineering Intern Nov 2024 – Feb 2025

- Contributing to full-stack development of **Align**, a mental wellness app using **Flutter** and **Python Flask**.
- Developing an **NLP-based journaling feature** that analyzes user entries with **TextBlob** to categorize emotional tone and generate summaries.
- Enhancing backend APIs and optimizing response latency for scalable deployment.

PROJECTS

Anime Recommendation System

- Built a personalized recommendation engine using **Spark MLlib (ALS + KMeans)** trained on a 17K-title dataset to cluster users by viewing patterns.
- Designed and developed a full **Flask + React web interface** with custom layouts using **HTML/CSS** for dynamic recommendation displays.
- Integrated an **SQLite backend** for persistent user data storage and real-time query retrieval.

Reddit Search Engine

- Developed a Reddit-based search engine using **PyLucene** and **Flask**, implementing the **BM25 ranking function** and NLP-based text preprocessing for query relevance.
- Deployed a simple front-end search interface for visualizing ranked results and comparing retrieval performance across algorithms.

GPU and CPU Maze Solver

- Implemented a **parallel BFS maze solver in CUDA** achieving up to **12× speedup** compared to CPU execution.
- Designed and benchmarked an equivalent **C++ sequential maze solver**, comparing runtime efficiency and documenting performance trends in a comprehensive report.
- Optimized **thread block sizing**, memory access patterns, and load balancing to maximize GPU throughput for 512×512 mazes.

Feature Selection for KNN

- Designed and implemented **forward and backward feature selection algorithms** in Python to optimize KNN accuracy on multidimensional datasets.
- Visualized feature importance and model performance through iterative validation results.

Automate 8 Puzzle Solver

- Developed an intelligent agent applying **heuristic search** and **game-tree algorithms** (Minimax, Alpha-Beta pruning) for optimal path planning and move prediction.
- Analyzed computational tradeoffs between heuristic evaluation functions and implemented metrics to track performance improvements.

TECHNICAL SKILLS

- **Languages:** Python (NumPy, Pandas, Scikit-Learn, TextBlob, Flask), C++, Java, HTML/CSS, JavaScript
- **Frameworks/Tools:** Spark MLlib, React, PyLucene, SQLite, Git, TailwindCSS, CUDA
- **Domains:** AI/ML, NLP, Recommendation Systems, Information Retrieval, Data Analytics, Software Engineering